Advantages of Nitriding

1. The surface hardness is very high.
2. It increases wear and corrosion resistance.
3. Due to rapid cooling there is no scaling.
4. It is suitable for complicated structures.

CYANIDING

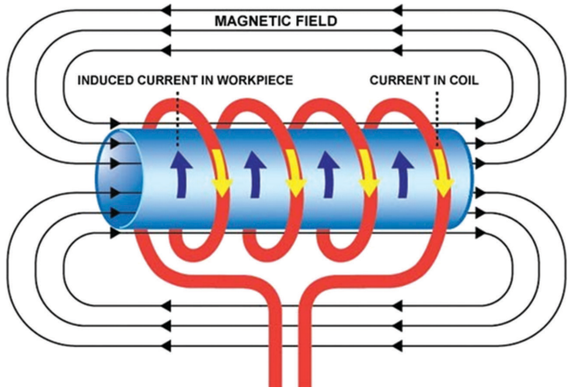
Cyaniding is one of the case hardening process on low carbon and alloy steels. The metal is heated at the range of temperature 800°C to 900°C in contact with molten bath of sodium cyanide, so the surface of the metal absorbs carbon and Nitrogen. Then the metal is quenched in oil or water, depending on the type of material used and the hardness required. In this process the surface is hardened upto a thickness of 0.5mm.

Advantages of Cyaniding

1. It is a rapid process.
2. The exterior will have high lustre.
3. It gives uniform hardening.
4. It increases the ductility.

INDUCTION HARDENING

It is a process used for the surface hardening of steel and other alloy components. The parts to be heat treated are placed in a induction coil. When high frequency alternating current is passed through the coil, the coil produces an alternative magnetic field within the components. So, the outer surface of the part is heated to higher critical temperature. Then the components are quenched in oil or another media, depending upon the types of steel and hardness desired.



Induction Hardening

Advantages

1. It is a rapid process.
2. High rate of change of properties.
3. The exterior surface can be made of high quality.
4. The internal hardening can be controlled.
5. Heat can be controlled in this process.

Disadvantages

1. The cost of the equipment is very high.
2. Low carbon steel cannot be hardened.
3. Maintenance cost of the equipment is more.

FLAME HARDENING

Flame hardening is a process of hardening the surface of the steel. Here, the steel is heated to the above critical temperature by using an oxy-acetylene flame and then it is quenched by spraying water. The result is a hard surface Layer on the steel. In this process the outer surface is hardened 3 to 6mm.

Advantages

1. It is simple and manufacturing cost is less.

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